

Microprocessor, microcontroller and embedded system

- ▶ Course code: CSE-3523
- ▶ Credit Hour: 3

Syllabus

#	Content	Duration	CLOs
Mid-Term (30 Marks)			
1	Introductory Concept: Evaluation of microprocessor, Types of microprocessor, system bus, hardware of a microprocessor, memory-addressing technique. An overview of Intel 80186, 80286, 80386, 80486 and Pentium microprocessor;	2 weeks	CLO1
2	8086 Microprocessor: properties, architecture, registers, FLAGS register, physical address calculation, addressing modes, Addressing Techniques of 8086 Microprocessor.	2 weeks	CLO1, CLO2
3	Instruction: Instruction set, Instruction format, Fetch-decode-Execution cycle.	2 weeks	CLO2
Final Exam: 50 Marks			
Group-A (20 Marks)			
4	Interrupt System: Sources of interrupt, Types of interrupt, handling interrupt request, interrupt vector and table, 8259A priority interrupt controller, Daisy chain.	2 weeks	CLO2
5	Advanced microprocessor: Embedded microprocessor, Bit slice microprocessor, arithmetic processor, Multitasking, Itanium and Merced Microprocessor, Pipelining, DMA technique,	2 weeks	CLO2
Group-B (30 Marks)			
6	I/O operation: Isolated and memory mapped I/O, Basic Input and Output Interfaces, Handshaking, Address decoding, 8255A Programmable peripheral Interface, I/O ports, I/O processor	1 week	CLO3
7	Microcontroller and Embedded Systems: Architecture of 8051, Signals, Operational features, Memory and I/O addressing, Interrupts, Instruction set, Applications, Introduction to Embedded Systems, Characteristics, Components, Hardware-Software Partitioning, Specification and Modeling, Optimizing Design, SOC, and Application Domains.	2 weeks	CLO3
8	IoT and its Technology: The Internet of Things basics, Infrastructure, Networks and Communication, Processes, Data Management, Trust M2M to IoT, M2M Value Chains, IoT Value Chains, IoT Architecture—State of the Art, Value Creation from Big Data and Serialization, Applications of IOT, privacy and security an overview towards Governance policy, First Steps Towards a Secure Platform, Smart Approach, Data Aggregation for the IoT in Smart Cities, and Security.	2 weeks	CLO3

Reference books

- ▶ **The Intel Microprocessors**
By Barry B. Brey
- ▶ **Microprocessors and Interfacing**
by Douglas V Hall
- ▶ **Microprocessor and Microcomputer based system design**
by MD. Rafiquzzaman

Segment 1: Introduction

A COMPUTER SYSTEM

- ▶ A computer is a programmable machine that receives input, stores and manipulates data/information, and provides output in a useful format.

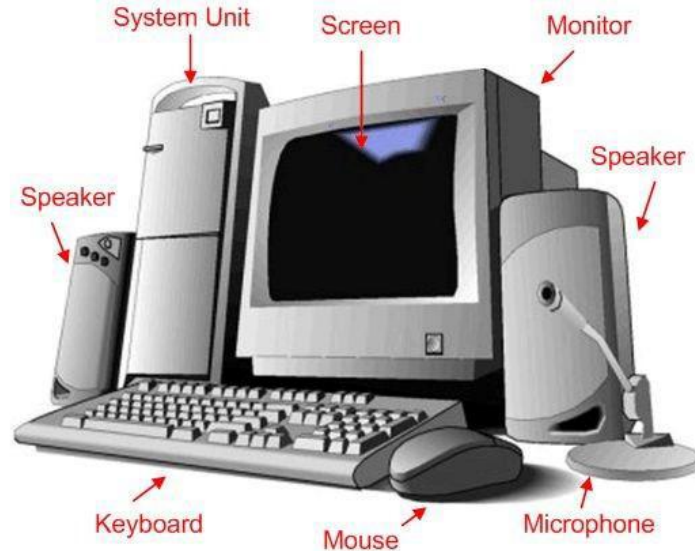


Diagram Of A Computer System

The Microprocessor

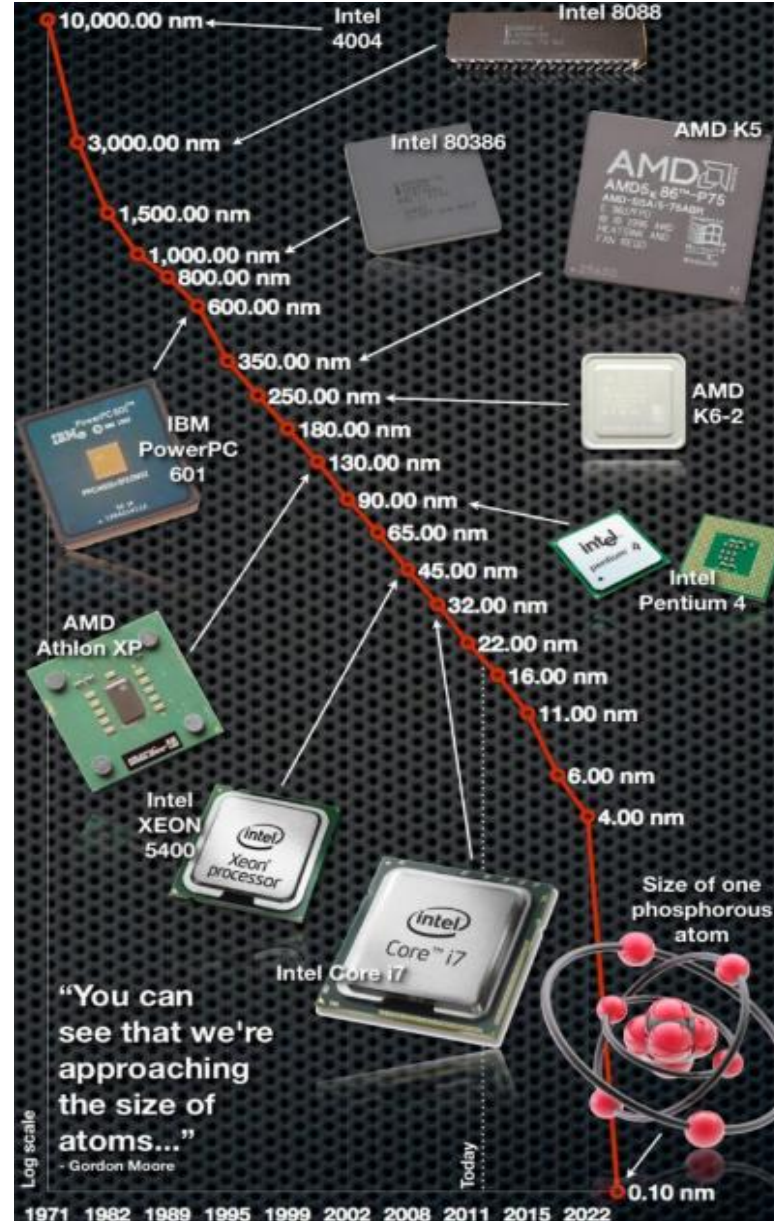
- ▶ Microprocessor, the **heart** of the microprocessor-based computer system **is an IC**.
- ▶ Microprocessor is the **controlling element** in a computer system.
- ▶ **It controls memory and I/O devices** through buses and transfers instructions and data between microprocessor and memory or I/O device.

Main Tasks of Microprocessor

- ❑ The microprocessor performs three main tasks for the computer system:
 - ▶ 1. **Data transfer** between microprocessor and memory or I/O devices.
 - ▶ 2. Simple **arithmetic and logic operations**.
 - ▶ 3. **Program flow** via simple decisions.

The power of the microprocessor is in its capability to execute billions of millions of instructions per second from a program or software (group of instructions) stored in memory.

Evolution of the Microprocessor



Evolution of The Microprocessor

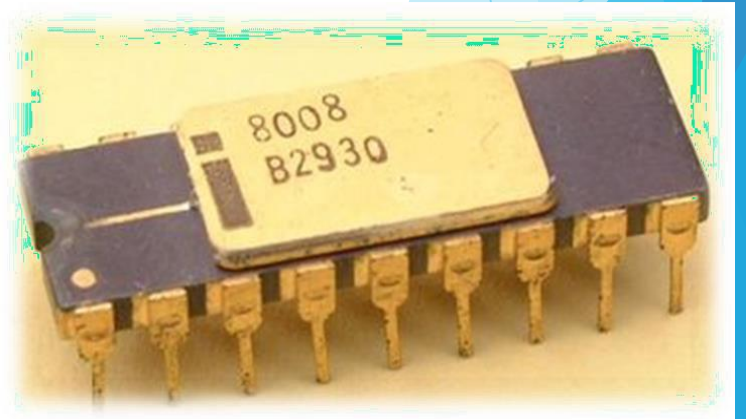
1st Microprocessor

- ▶ 1971, INTEL corp., 4004
- ▶ 4-bit microprocessor
- ▶ 4-bit memory, contain only 45 instructions
- ▶ Fabricated with P-channel MOSFET
- ▶ Execute instructions at the slow rate of 50 KIPs (Kilo- instructions per second)



2nd Microprocessor

- ▶ Later in 1971, INTEL corp., 8008
- ▶ 8-bit microprocessor
- ▶ 16kB memory, contain only 48 instructions,
- ▶ Execution time 20 micro-seconds



Evolution of The Microprocessor

3rd Microprocessor (1st modern microprocessor)

- ▶ 1973, INTEL 8080
- ▶ 8-bit microprocessor
- ▶ 64kB memory
- ▶ TTL technology
- ▶ Execution time 2 micro-seconds



- The 8085 Microprocessor

- 1977, Intel 8085
- 8-bit microprocessor (Data bus 8 bits, address bus 8 bits)
- 64kB memory
- Execution time 1.3 microseconds

Advantage:

1. internal clock generator
2. Internal system controller
3. Higher clock frequency (3MHz)



Evolution of The Microprocessor

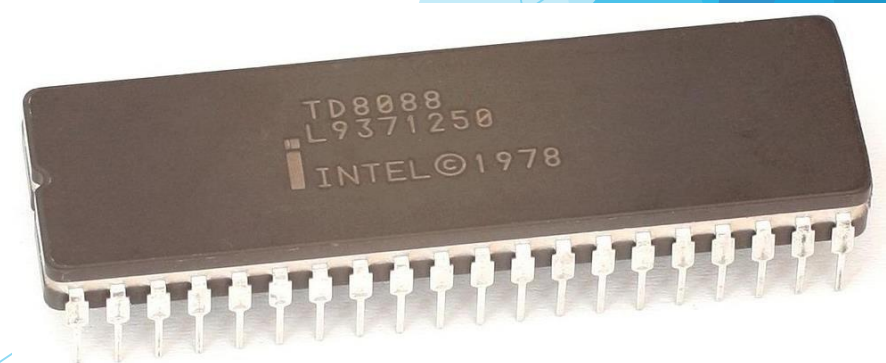
- **The 8086 Microprocessor**

- ▶ 1978, Intel 8086
- ▶ 16-bit microprocessor (Data bus 16 bits, address bus 20 bits)
- ▶ 1MB memory
- ▶ Execution time 400 nano-seconds
- ▶ Clock frequency 5MHz
- ▶ Cache or queue memory 6 byte



- **The 8088 Microprocessor**

- ▶ 1979, Intel 8088
- ▶ Same as 8086 but speedy than 8086



Evolution of The Microprocessor

The 80286 Microprocessor

- ▶ 1983, INTEL 80286
- ▶ 16-bit microprocessor
- ▶ 16MB memory
- ▶ Execution time 250 nano-seconds
- ▶ Clock frequency 8MHz

The 80386 Microprocessor (1st practical 32-bit MP)

- ▶ 1986, Intel 80386
- ▶ 32-bit microprocessor (Data bus 32 bits, address bus 32 bits)
- ▶ 4GB memory
- ▶ Cache or queue memory 8kB

Have few modified version (80386SX, 80386SL/ 80386SLC, 80386EX



Evolution of The Microprocessor

- The 80486 Microprocessor
 - ▶ 1989, INTEL 80486
 - ▶ Same as 80386
 - ▶ Execution time 25 nano-seconds
 - ▶ Clock frequency 50MHz



Evolution of The Microprocessor...

The Intel Pentium Microprocessor

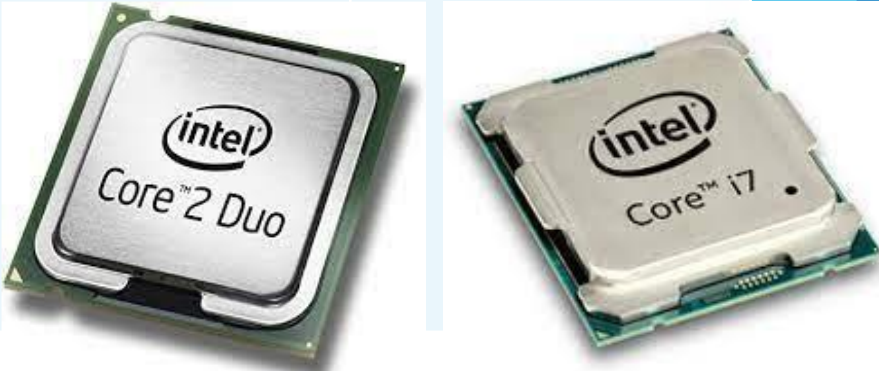
- ▶ 1993
- ▶ 64-bit microprocessor
- ▶ 4GB memory
- ▶ Cache or queue memory 16kB
- ▶ Pentium contained 8KB instruction cache & 8KB data cache
- ▶ Execute instructions at the rate of 110 MIPS
- ▶ Two version of Pentium operated with a clocking frequency of 60MHz & 66MHz
- ▶ Double-clock Pentium, operating at 120MHz & 133MHz also available



History of Microprocessor by INTEL

Processor	Year of Introduction	Numbers of Transistors	Clock Speed	Address bus	Data Bus	Addressable Memory
4004	1971	2300	108kHz	10bits	4bits	640Bytes
8008	1972	3500	200kHz	14bits	8bits	16K
8085	1976	6500	5MHz	16bits	8bits	64K
8086	1978	29000	5MHz	20bits	16bits	1M
80386	1985	275000	16MHz	32bits	32bits	4G
80486	1989	1.2M	25MHz	32bits	32bits	4G
P1 – P4	1995-2000	3.1M-42M	150MHz-1.4GHz	32bits-36bits	32bits-64bits	4G-64G

Evolution of The Microprocessor...

Name	Year of Invention	Clock speed (GHz)	Number of transistor	Inst. per sec
INTEL core 2	2006 (other versions core2 duo, core2 quad, core2 extreme)	1.2 to 3	291 Million transistors	64K KB of L1 cache per core, 4 MB of L2 cache
i3, i5, i7	2007, 2009, 2010	2.2-3.3, 2.4-3.6, 2.93-3.33		

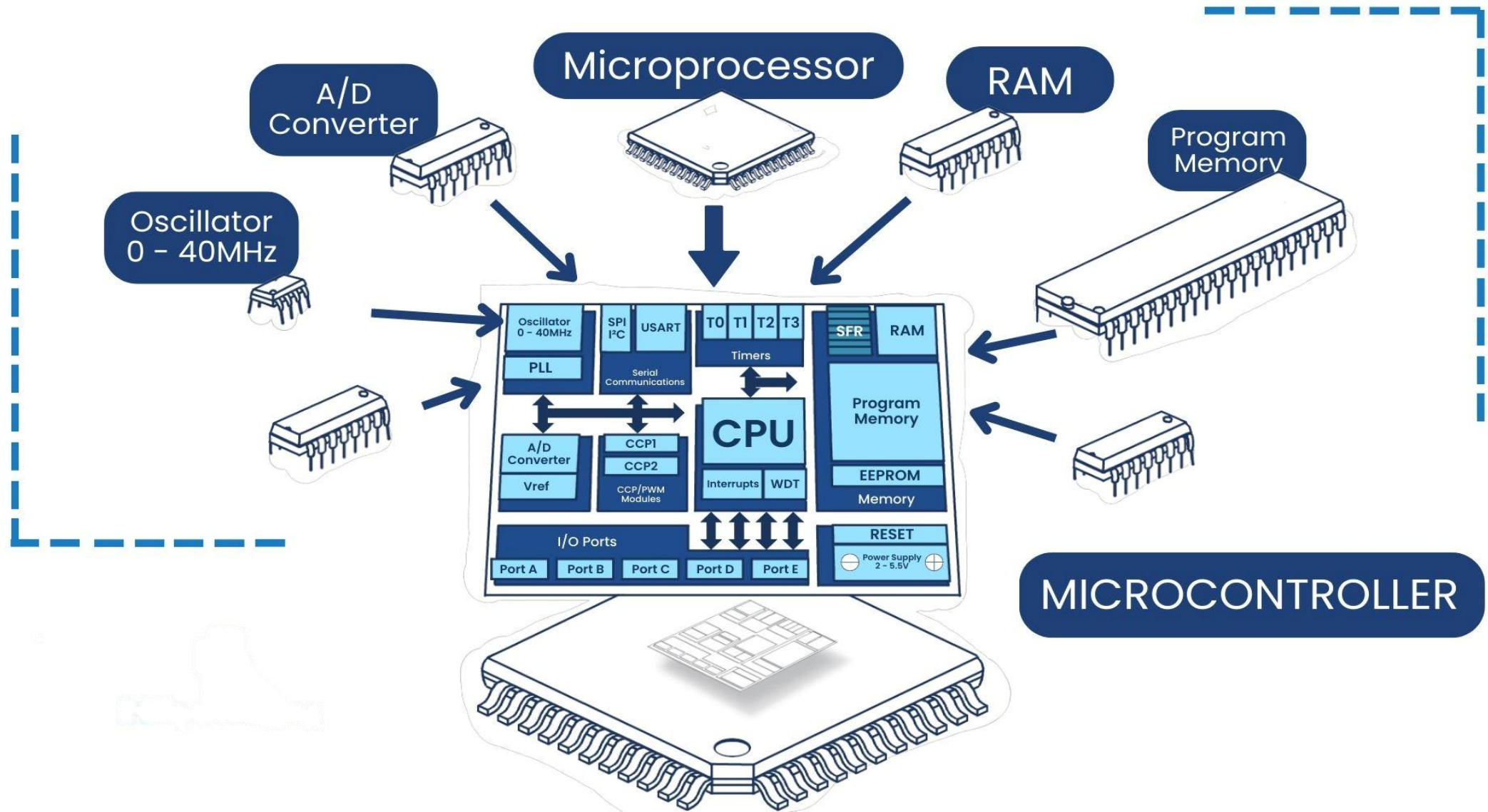
❑ Fifth-generation of MP

- From 1995 to now, we are in the fifth generation. 64-bit processors like Pentium, Celeron, dual, quad, and octa-core processors came into existence.

Microcontroller

- ▶ A device/microcomputer that includes **a microprocessor, memory, and several signal lines** on a single chip fabricated using VLSI technology. A single-chip microcomputer is called a microcontroller.
- ▶ The microcontroller on single chip, which typically includes a microprocessor, 64 byte RAM, 1-2 kilo byte ROM and several signal lines to connect I/O devices.
- ▶ Microcontrollers are used for such as controlling appliances, traffic lights, etc.
- ▶ Typical IC's example are Zilog-Z8, Intel MC551, Motorola 68HC11.

Microcontroller



Microprocessor vs MicroController

Microprocessor	Microcontroller
Microprocessor is the heart of Computer system.	MicroController is the heart of an embedded system.
It is only a processor, so memory and I/O components need to be connected externally	MicroController has a processor along with internal memory and I/O components.
Memory and I/O have to be connected externally, so the circuit becomes large.	Memory and I/O are already present, and the internal circuit is small.
You can't use it in compact systems	You can use it in compact systems.
Cost of the entire system is high	Cost of the entire system is low

Generations of microprocessors:

First generation

- ▶ From **1971 to 1972** the era of the first generation came which brought microprocessors like INTEL 4004 Rockwell international PPS 4 INTEL 8008 etc

Second generation

- ▶ The second generation marked the development of **8 bit** microprocessors **from 1973 to 1978** Processors like INTEL 8085 Motorola 6800 and 6801etc came into existence

Third generation

- ▶ The third generation brought forward **16 bit** processors like INTEL 8086 80186 80286 Motorola 68000 68010 etc **From 1979 to 1980** this generation used HMOS technology

Generations of microprocessors:

Fourth generation

- ▶ The fourth generation came into existence from **1981 to 1995** The 32 bit processors using HMOS fabrication came into existence INTEL 80386 and Motorola 68020 are some of the popular processors of this generation

Fifth generation

- ▶ From **1995 till now** we are in the fifth generation **64 bit** processors like PENTIUM, Celeron, dual, quad, and octa core processors came into existence

Some Important Terms

- ▶ Bit (b) - a binary digit, 0 or 1.
- ▶ Byte (B) - a group of 08 bits. bits.(1 byte 8 bits)
- ▶ Word (W) - a group of 02 bytes. bytes.(1 word=2 bytes=16 bits)
- ▶ Instruction- a command in binary recognized executed by the computer.

Microprocessor-based Computer System

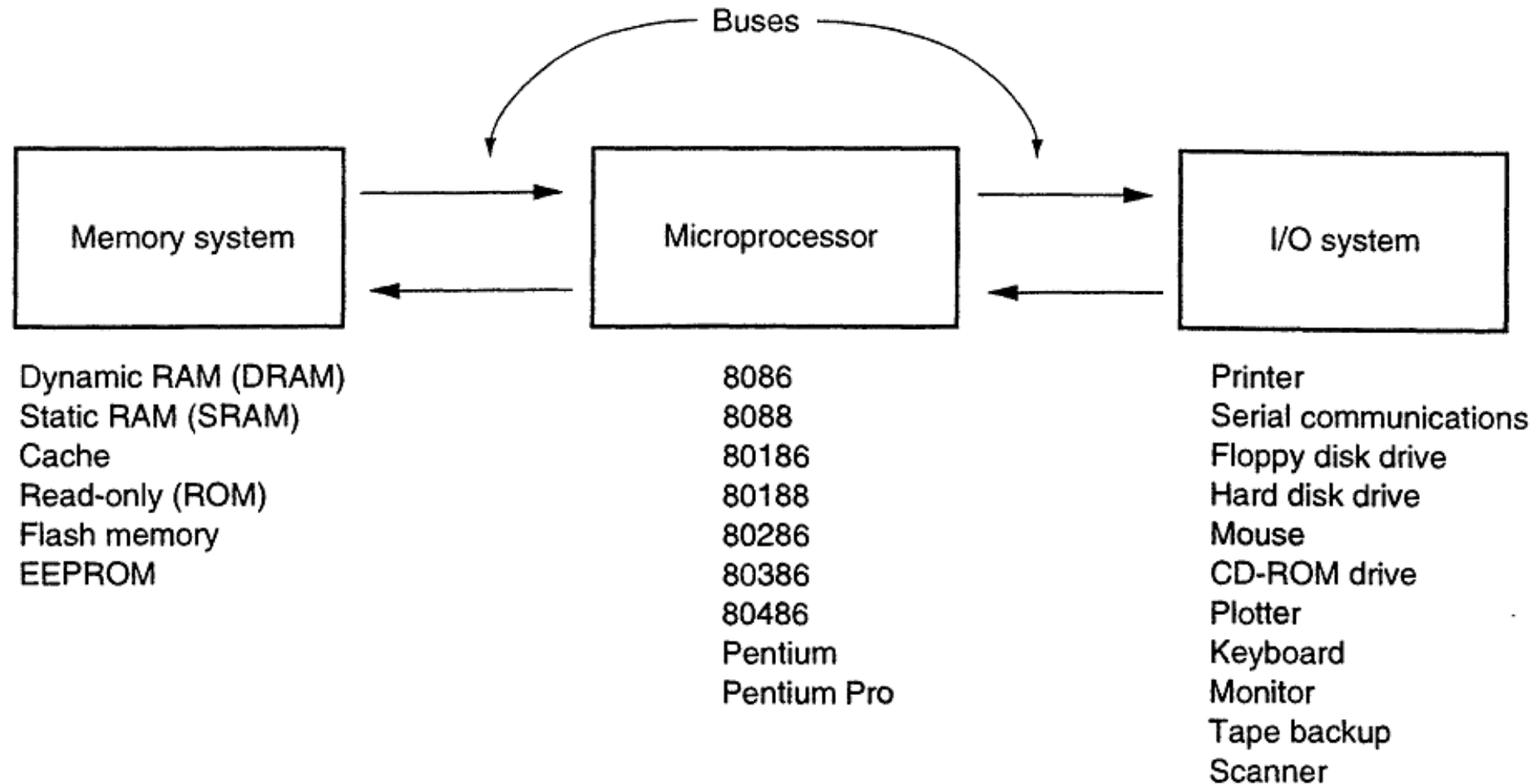


FIGURE 1-2 The block diagram of a microprocessor-based computer system

Types of Microprocessor

Complex Instruction Set Computer (CISC)

- ▶ CISC or Complex Instruction Set Computer is a computer architecture where instructions are such that a **single instruction can execute multiple low-level operations** like loading from memory, storing into memory, or an arithmetic operation, etc. Example:
- ▶ Example: 1. Intel 386 2. Intel 486 3. Pentium

Reduced Instruction Set Computer (RISC)

- ▶ RISC or Reduced Instruction Set Computer is a computer architecture where **instruction is simple and designed to get executed quickly.**
- ▶ Example: 1. IBM RS6000 2. MC88100 3. DEC Alpha 21064

Types of Microprocessor

Explicitly Parallel Instruction Computing (EPIC)

- ▶ EPIC or Explicitly Parallel Instruction Computing permits **computers to execute instructions parallel using compilers**. It allows complex instructions execution without using higher clock frequencies. Example:

1. Superscalar microprocessor

- ▶ These processors **can perform many tasks at a time**. They can be used for ALUs and multiplier-like arrays. They have multiple operation units and perform tasks by executing multiple commands.

2. Application-specific integrated circuit

- ▶ These processors are application-specific, like personal digital assistant computers. They are designed according to proper specifications

Characteristics of a microprocessor

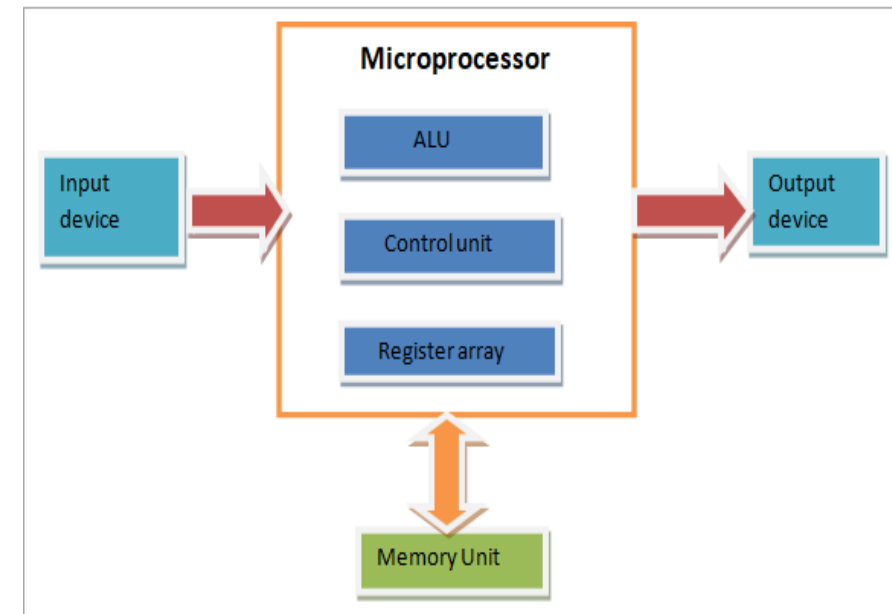
- ▶ 1. It assimilates the functions of a **CPU** on to a single IC.
- ▶ 2. It can perform some basic operations like addition, subtraction, multiplication, division and some logical operations using its **Arithmetic and Logical Unit (ALU)**.
- ▶ 3. **Move data** from one location to another.
- ▶ 4. It has a program counter that can **store the addresses** of next instruction.
- ▶ 5. Computational capacity is very high.
- ▶ 6. Perform the instruction cycle like as **fetch, decode and execute** repeatedly.
- ▶ 7. To design a system with a microprocessor, it needs to connect with **RAM, ROM, I/O devices** etc.
- ▶ 8. Building cost is high.
- ▶ 9. **Power consumption** and dissipation is high.

Characteristics of a microprocessor

- ▶ **Three basic characteristics** that differentiate microprocessors are-
- ▶ • **Instruction set:** The set of instructions that the microprocessor can execute. MIPS stands for "millions of instructions per second" and is a rough measure of the performance of a CPU.
- ▶ • **Bandwidth or Data Width:** **The number of bits processed in a single instruction.** Data Width is the width of the ALU. An 8-bit ALU can add/subtract/multiply/etc. two 8-bit numbers, while a 32-bit ALU can manipulate 32-bit numbers.
- ▶ • **Clock speed:** Given in megahertz (MHz), **it determines how many instructions per second the processor can execute.** For example, a 32 bit microprocessor that runs at 50MHz is more powerful than a 16-bit microprocessor that runs at 25MHz.

Hardware of a microprocessor

- ▶ **Arithmetic Logic Unit (ALU)**
Performs arithmetic & logical operations.
- ▶ **Control Unit (CU)** -Directs and manages instruction execution.
- ▶ **Registers**-Small storage locations for quick data access.
- ▶ **Buses (Data, Address, Control)** -Transfers data and signals between components



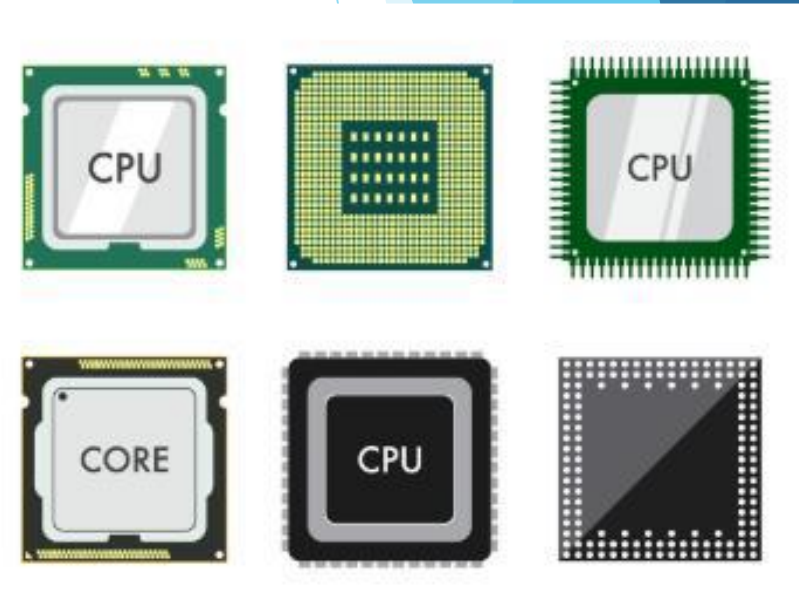
Central Processing Unit (CPU)

Central Processing Unit (CPU) -The Brain of the Computer

- ▶ •The CPU controls and executes all computer tasks.
- ▶ •It performs arithmetic & logical operations efficiently.
- ▶ **Main Components:**
- ▶ •ALU (Arithmetic Logic Unit): Handles calculations & logic.
- ▶ •CU (Control Unit): Directs and manages operations.

▶ How It Works?

- ▶ 1.Receives input from the user.
- ▶ 2.Processes data using ALU & CU.
- ▶ 3.Generates and delivers output. The CPU is the heart of a computer, ensuring smooth and efficient functioning of all operations.

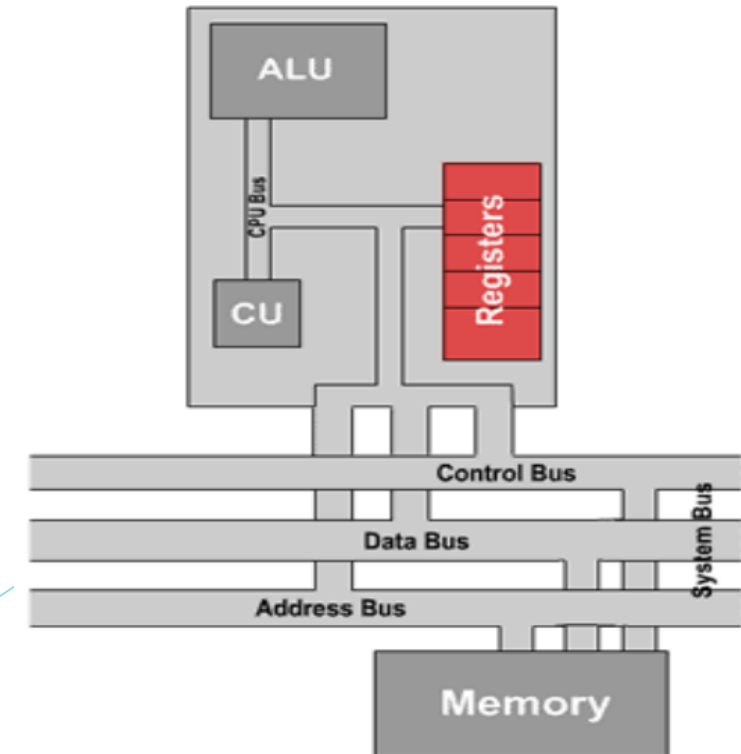
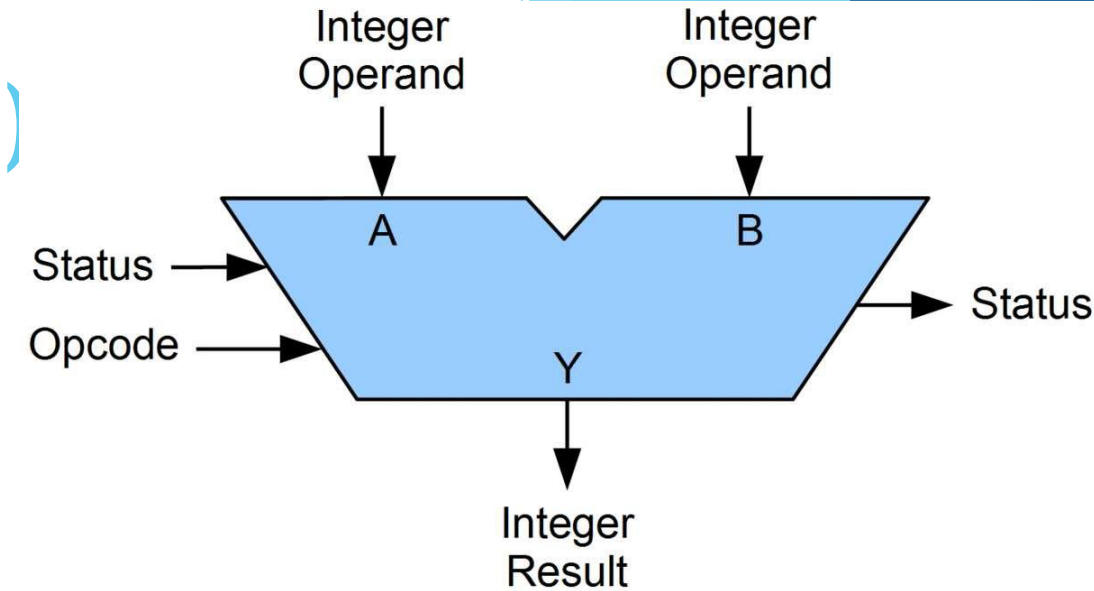


Arithmetic & Logic Unit (ALU)

i. Arithmetic & Logic Unit (ALU)

□ Some of the tasks performed by the ALU are given below:

- ▶ Addition and subtraction
- ▶ Multiplication and division
- ▶ Logical tests
- ▶ Comparison
- ▶ Bit shifting



Control Unit (CU)

- ▶ **Controls the movement of instructions** in and out of the processor, and also controls the operation of the ALU..
- ▶ The **three main** elements of the control unit
 1. **Decoder:** This is used to decode the instructions that make up a program when they are being processed.
 2. **Timer or clock:** The timer or clock ensures that all processes and instructions are carried out and completed at the right time
 3. **Control logic circuits:** **used to create the control signals** themselves, which are then sent around the processor..

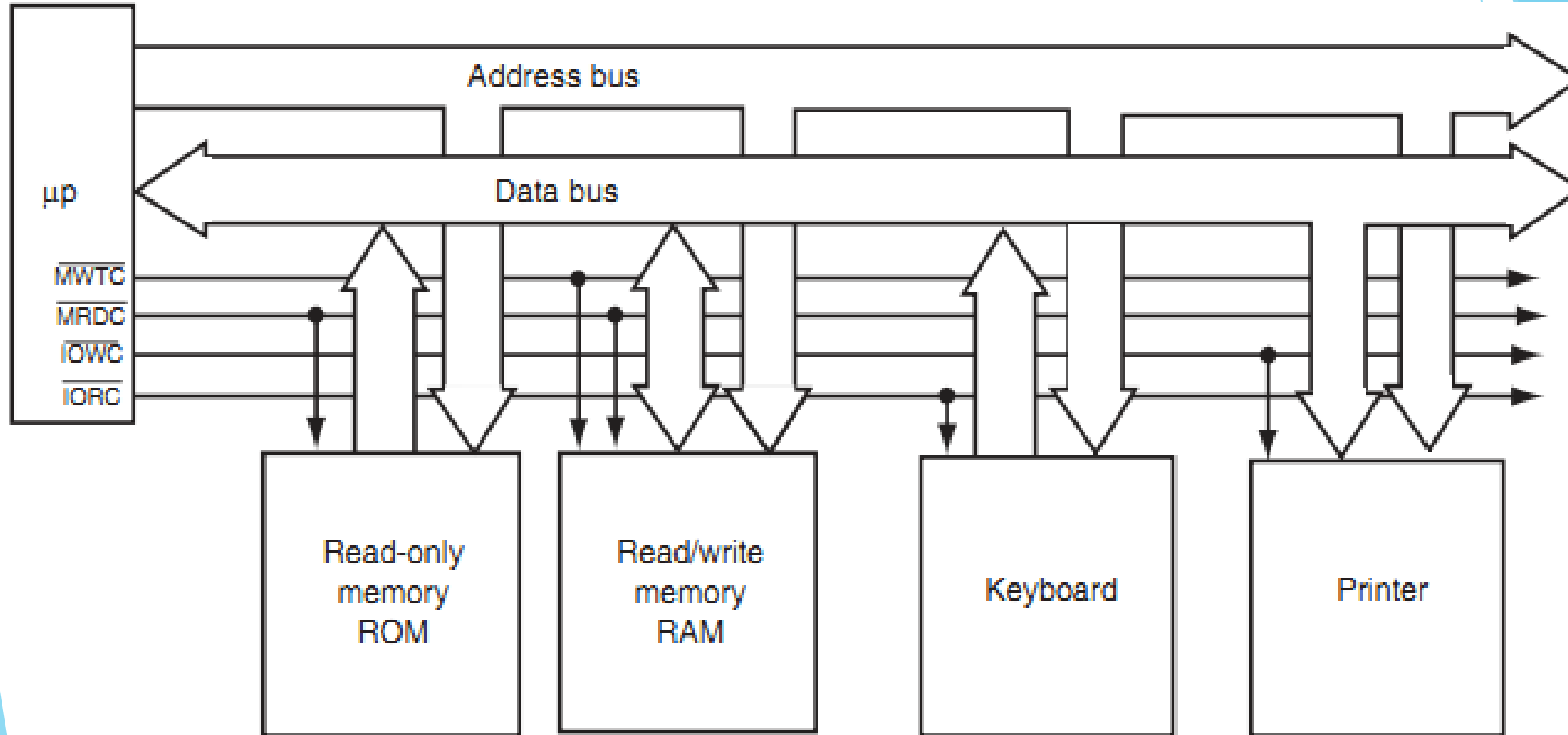
Memory Array Register

- ▶ Some common registers used for specific functions. These are:
- ▶ ☐ Program Counter (PC): used to hold the memory address of the next instruction that has to execute in a program.
- ▶ ☐ Instruction Register (IR): used to hold the current instruction in the processor while it is being decoded and executed.
- ▶ ☐ Accumulator (A, or ACC): used to hold the result of operations performed by the arithmetic and logic unit.
- ▶ ☐ Memory Address Register (MAR): Used for storage of memory addresses, usually the addresses involved in the instructions held in the instruction register

Memory Array Register

- ▶ **Memory Buffer Register (MBR):** When an instruction or data is obtained from the memory or elsewhere, it is first placed in the memory buffer register.
- ▶ **Flag register/status flags:** The flag register is specially designed to contain all the appropriate 1-bit status flags,
- ▶ **Other general-purpose registers:** generally used for the quick storage of pieces of data that are required later in the program execution..

Buses:system Bus



Buses....

Block diagram of bus structure

***Write-MP to others
Read-others to MP***

Buses: System Buses

- ▶ A group of lines used to **transfer bits between microprocessor and other components** (RAM, ROM, I/O) of computer is called bus

- ▶ There are **3 types** of System buses

1.Data bus(Bi-directional)

- ▶ Transfers information between microprocessor and its memory and I/O address space.

2.Address bus(Unidirectional)

- ▶ Requests a memory location from the memory or I/O location from I/O devices.

Buses: system Buses

- ▶ 3.Control bus(Unidirectional)
- ▶ Contains lines that **select memory or I/O** and cause them to perform a read or write operation.

Four control bus connections:

- ▶ MRDC-Memory read control
- ▶ IORC-I/O read control
- ▶ MWTC-Memory write control
- ▶ IOWC -I/O write control

Function of Buses....

- ▶ •**1.Data sharing:** All types of buses found on a computer must be able to transfer data between the computer peripherals connected to it.
- ▶ •**2. Addressing:** A bus has address lines, which match those of the processor. This allows data to be sent to or from specific memory locations.
- ▶ •**3.Power:** A bus supplies power to various peripherals that are connected to it.
- ▶ •**4.Timing:** The bus provides a system clock signal to synchronize the peripherals attached to it with the rest of the system.

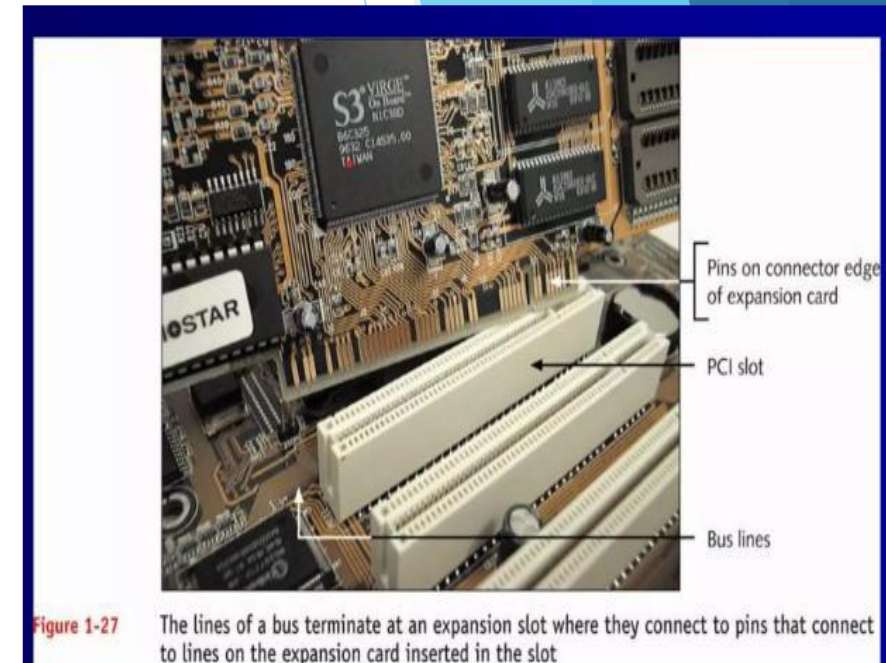
Expansion Type Buses....

Integrated Motherboards

- ▶ •Common devices (keyboard, mouse, audio, video, LAN) are built-in.
- ▶ •Lower overall cost, simpler design.
- ▶ •Expensive to repair, limited customization.
- ▶ •Not ideal for high-performance applications like gaming or advanced graphics.

Expansion Cards

- ▶ •Used for specialized devices requiring higher performance.
- ▶ •Installed in expansion slots on the motherboard.
- ▶ •Examples: Graphics Cards, Sound Cards, Network Cards.



Expansion Type Buses....

- ▶ These are some of the common expansion bus types that have ever been used in computers
- ▶ •**PCI** -Peripheral Component Interconnect
- ▶ •**PCMCIA** -Personal Computer Memory Card Industry Association (Also called PC bus)
- ▶ •**AGP** -Accelerated Graphics Port
- ▶ •**SCSI**-Small Computer Systems Interface

Peripheral Component Interconnect (PCI)

- ▶ •Latest development in bus architecture, standard for expansion cards.
- ▶ •Connects CPU, memory, and peripherals for faster data transfer.
- ▶ •Supports 32-bit & 64-bit data width, compatible with 486s & Pentiums

Expansion Type Buses....

Accelerated Graphics Port (AGP)

- ▶ Developed for high-performance video processing.
- ▶ Directly connects to CPU, operating at processor bus speed.
- ▶ Uses main PC memory to store 3D images, allowing unlimited video memory.

PC Card

- ▶ Developed by PCMCIA for laptop computers.
- ▶ Standard bus used in small computers.

Small Computer System Interface (SCSI)

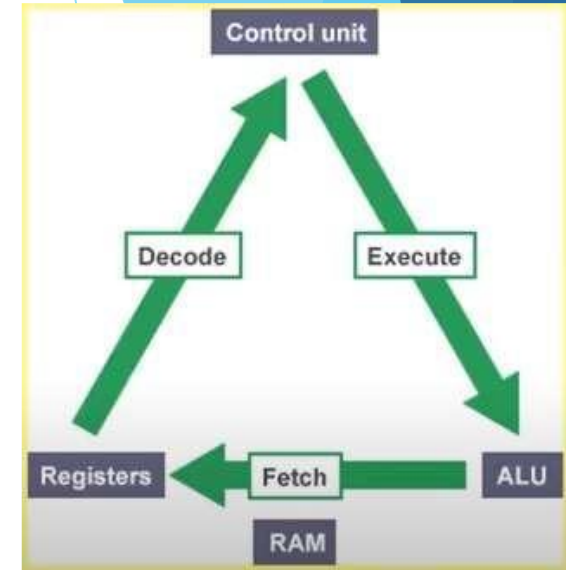
- ▶ Parallel interface standard for Apple, PC & Unix systems.
- ▶ Used for attaching peripheral devices efficiently.

How does a Microprocessor Work?

- ▶ The Microprocessor follows a sequence: Fetch, Decode, Execute

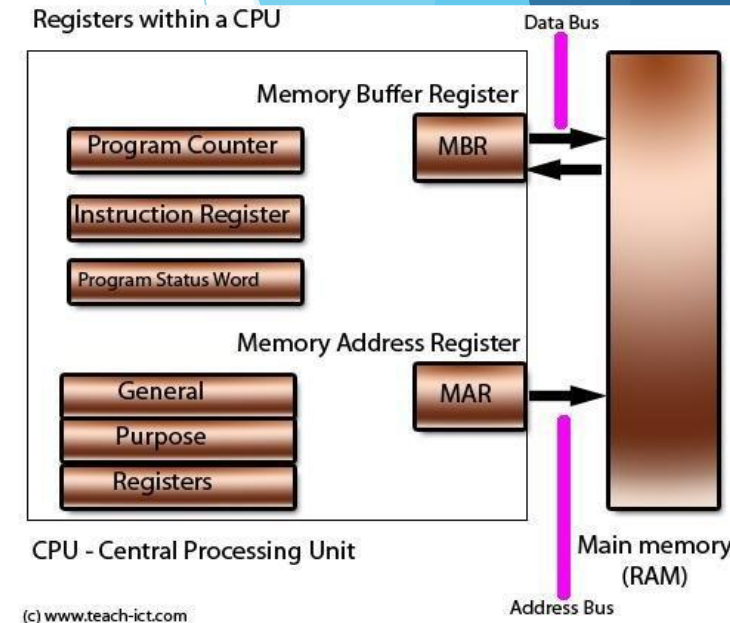
Steps in the Execution Cycle:

- ▶ **Fetch:** The microprocessor retrieves an instruction from memory.
- ▶ **Decode:** It interprets the instruction to determine the required operation.
- ▶ **Execute:** The Arithmetic Logic Unit (ALU) performs computations.



Fetch

- ▶ **Step 1:** The Program Counter copies the address of the next instruction it contains into the Memory Address Register (MAR).
- ▶ **Step 2:** The Memory Address Register places the address to be used on to the 'Address Bus'
- ▶ **Step 3:** The Memory Address Register triggers a 'read' signal that causes main memory (RAM) to place the instruction being asked for on to the 'Data Bus'
- ▶ **Step 4:** The instruction on the data bus is loaded into the Memory Data Register (also called Memory Buffer Register)
- ▶ **Step 5:** The Memory Data register copies the instruction into the 'Instruction Register' The Fetch stage is now complete



Decode and Execute

Decode:

- ▶ •The Control unit examines the instruction in the current instruction register (CIR) and 'decodes' it. This means a special part of Control unit called the 'decode' unit will make the rest of the CPU ready to carry out the instruction.
- ▶ •For example the instruction might say 'Add'. The decode unit understands what this means and gets the system ready to carry out that instruction.

Execute:

- ▶ The instruction within the instruction register is carried out (executed) by the ALU. The part that executes instructions is called the 'execute unit'.

Memory-address

What is a Memory Address?

- A memory address is simply the **unique number (location)** assigned to **each byte stored in memory**.
 - Since memory in computers is **byte-addressable**, each byte has a separate address.
 - A program uses these addresses to **store, fetch, or modify data**.
-
- ▶ If memory starts at address 00000H, the next byte will be at 00001H, then 00002H, and so on
 - ▶ If memory size = M bytes, then the required **address size (N bits)** is: $N = \log_2 M$

Examples

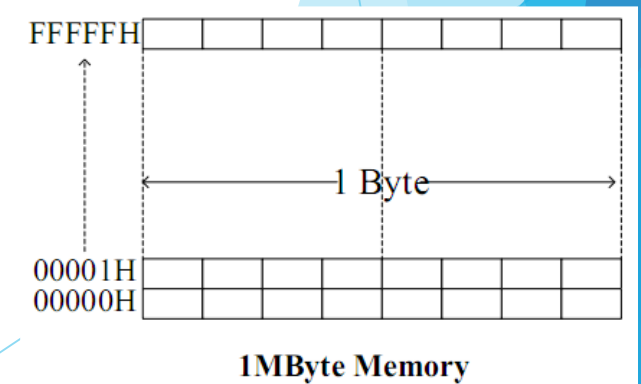
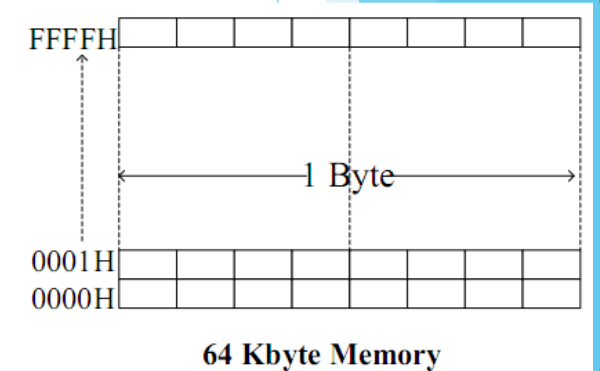
- ▶ 64 KB memory = $64 \times 1024 = 65536 \rightarrow \log_2(65536) = 16$ address lines.
- ▶ 1 MB memory = 1,048,576 bytes $\rightarrow \log_2(1048576) = 20$ address lines.
- ▶ 4 GB memory = 32 address lines.

Memory-address

- **How many bits are necessary to address 1MB memory?**

- ▶ 1MB= 1024Kbyte= 1,048,576 Byte = 2^{20} Byte. So, we need 20 bits to address 1MByte memory. Addresses will be-
- ▶ 0 to 1,048,575 (Decimal)
- ▶ 00000H to FFFFFH (Hex)
- ▶ Lowest value: 0000 0000 0000 0000 0000 - (00000H)
- ▶ Highest value 1111 1111 1111 1111 1111 - (FFFFFFH)
- ▶ N.B. 8086 has 20-bit address bus. So, it can addresses 1 MB memory

- For addressing 64 KB of memory, we need at least 16 bits.



Intel Processor Generations and Architectures

- ▶ •Intel 80186 -Foundation for Embedded Systems
- ▶ •Intel 80286 -Enter Protected Mode
- ▶ •Intel 80386 -32-bit Breakthrough
- ▶ •Intel 80486 -Speed and Integration
- ▶ •Intel Pentium -Superscalar Power

Intel 80186 -Foundation for Embedded Systems

- ▶ Introduction Year:1982
- ▶ Architecture:16-bit processor
- ▶ Clock Speed : MHz to 25 MHz
- ▶ Address Bus:20-bit (can address up to 1 MB memory)
- ▶ Data Bus:16-bit

Key Features:

- ▶ •Enhanced instruction set over 8086, allowing faster and more efficient execution
- ▶ •Integrated system components: interrupt controller, timers, DMA controller,
- ▶ •Operates in **real mode only**(no protected mode like the 80286)
- ▶ •Optimized for **embedded systems** rather than desktop PCs

The Intel 80186 was designed as an upgraded version of the 8086, primarily targeting embedded applications. It featured integrated support hardware, which reduced overall system complexity and cost. Although not widely adopted in personal computers, it became popular in embedded devices such as printers, networking hardware, and industrial control systems.

Intel 80286 -Enter ProtectedMode

- ▶ **Introduction Year:**1982
- ▶ **Architecture:**16-bit processor with some 24-bit addressing
- ▶ **Clock Speed:**6 MHz to 25 MHz
- ▶ **Address Bus:**24-bit (can address up to 16 MB memory)
- ▶ **Data Bus:**16-bit
- ▶ **Key Features:**
 - ▶ •Protected mode support, allowing multitasking and better memory management
 - ▶ •Real mode compatible with 8086 software
 - ▶ •No built-in floating point unit (FPU), but could be used with external 80287 coprocessor
 - ▶ •Improved performance over 8086 with pipelining and better instruction execution

The 80286 was a major step from 8086, introducing protected mode to support advanced OS features and larger memory addressing, making it suitable for business applications and early multitasking systems

Intel 80386 -32-bit Breakthrough

- ▶ **Introduction Year:**1985
- ▶ **Architecture:**32-bit processor
- ▶ **Clock Speed:**12 MHz to 40 MHz
- ▶ **Address Bus:**32-bit (can address up to 4 GB memory)
- ▶ **Data Bus:**32-bit
- ▶ **Key Features:**
 - ▶ •Full 32-bit internal and external architecture
 - ▶ •Supports real mode, protected mode, and virtual 8086 mode
 - ▶ •Built-in Memory Management Unit (MMU)
 - ▶ •Supports multitasking and advanced operating systems
 - ▶ •Optional integrated floating point unit (80387 coprocessor)

The 80386 was Intel's first 32-bit CPU for PCs, vastly increasing memory and processing power, enabling more powerful software and operating systems like Windows NT and UNIX

Intel 80486 -Speed and Integration

- ▶ **Introduction Year:**1989
- ▶ **Architecture:**32-bit processor
- ▶ **Clock Speed:**20 MHz to 100 MHz (depending on model)
- ▶ **Address Bus:**32-bit (4 GB memory addressing)
- ▶ **Data Bus:**32-bit

Key Features:

Integrated FPU (Floating Point Unit):

- ▶ First Intel CPU to include a built-in math co-processor (in DX versions).
- ▶ Greatly improved the performance of scientific, engineering, and multimedia tasks.

On-Chip Cache (8 KB):

- ▶ Included a small but fast **Level 1 (L1) cache**, storing frequently accessed data.
- ▶ Reduced the need to fetch data from slower main memory.

Intel 80486 -Speed and Integration

Instruction Pipelining:

- ▶ Could start a new instruction before the previous one finished.
- ▶ This "pipeline" increased instruction execution speed.

Improved Bus Interface and Memory Handling:

- ▶ Supported faster memory access.
- ▶ Worked efficiently with 80386-compatible software and operating systems.

Variants:

- ▶ **80486SX:** Cheaper version without FPU.
- ▶ **80486DX:** Full version with FPU.
- ▶ **80486DX2/DX4:** Used internal clock multiplication for faster performance (e.g., DX2 ran internally at 66 MHz even if external clock was 33 MHz).

Common Uses:

- ▶ Personal computers (early Windows systems)
- ▶ Office applications and early multimedia
- ▶ Academic and business environments needing moderate computing power

Intel Pentium -Superscalar Power

- ▶ **Introduction Year:**1993
- ▶ **Clock Speed:**60 MHz to 3+ GHz (varies by generation)
- ▶ **Data Bus:**64-bit
- ▶ **Address Bus:**32-bit (can access up to 4 GB memory)
- ▶ **The Intel Pentium processor** introduced **superscalar design**, enabling parallel execution of instructions and significantly outperforming earlier CPUs like the 486. It was the first mainstream Intel CPU to use a **64-bit data bus** and came with an integrated **FPU**, making it suitable for advanced multimedia, gaming, and productivity software.

Key Features:

- ▶ **Superscalar Design:** Executes 2 instructions per clock cycle (U & V pipelines)
- ▶ **Integrated FPU:** Faster floating-point operations
- ▶ **On-chip Cache:**L1 cache (8 KB instruction + 8 KB data); L2 added later
- ▶ **MMX Technology:** Accelerated multimedia processing (Pentium MMX & later)
- ▶ **Branch Prediction & Pipelining:** Improves performance